Table of Contents

1. Introduction: 3

1.1 Overview of the System 3

1.2 Significance and Motivation 3

1.3 Project Contribution: 4

2. Problem Statement 5

2.1 Addressing Key System Gaps 5

2.2 Challenges in Existing Payroll Systems 5

2.3 Efficient Payroll Management with Automation 5

3. Objectives 5

4. Methodology 6

4.1 Requirement Identification 6

4.1.1 Literature Review 6

4.1.2 Requirement Analysis 6

4.2 Design Diagrams 6

4.2.1 UML Class Diagram 7

4.2.2 Use Case Diagram 8

4.2.3 Entity Relationship Diagram (ERD) 9

4.2.4 Activity Diagram 10

4.2.5 Sequence Diagram 11

4.2.6 Data Flow Diagram 12

4.2.4 Activity Diagram 13

5. Implementation 14

5.1 Employee Login Portal: 14

5.2 Admin Dashboard & Analytics: 14

5.3 New Employee Onboarding Form: 15

5.4 Employee Attendance Log & Management: 16

5.5 Administrator Leave Request Dashboard 16

5.6 Salary Structure Management: 17

5.7 Edit Employee Salary Form: 17

5.8 Bonus & Deduction Management: 18

5.9 Payroll Management & History 18
5.10 Payroll Generation Form: 19
5.11 Employee Profile & Attendance Overview: 19
5.12 Leave Request Management & Status Tracking: 20
5.13 Downloadable Employee Pay slip: 20
5.14 Detailed Employee Pay slip: 21
5.15 Employee Profile Update Form: 22
5.16 Employee Leave Request Form: 23
5.17 Employee Pay slip History: 23
6. Conclusion 24

1. Introduction:

1.1 Overview of the System

The Automated Payroll System is a web application designed to simplify and digitize payroll management in organizations. This system aims to automate key payroll tasks like salary calculations, tax deductions, attendance tracking, and pay slip generation. It uses Laravel for the backend and PHP for the frontend. Laravel offers a secure and modular structure that supports quick development and easy maintenance, while PHP creates a flexible and responsive user interface. The system includes separate modules for administrators and employees. Administrators manage employee data, define salary structures, approve leaves, and start payroll generation. Employees can log into their accounts to view attendance records, check salary details, and download pay slips directly from the system. The system automatically calculates gross and net pay based on attendance, overtime, and applicable deductions or tax rules. It generates pay slips. All payroll records are stored securely in the database for auditing and reporting. The Automated Payroll System eliminates manual calculations, reduces human errors, and saves valuable administrative time. It gives organizations a modern, paperless, and precise solution to improve payroll management efficiency.

1.2 Significance and Motivation

In many organizations, payroll management is still done manually or with spreadsheets. This can lead to issues like calculation mistakes, data duplication, delayed salary payments, and lack of transparency. The push for developing this project came from the need to automate the payroll process to ensure accuracy, transparency, and efficiency. The system aims to lessen the workload for payroll officers and ensure that every employee is paid accurately and on time. With Laravel's built-in security and database features, this system also maintains data safety and scalability. Additionally, the motivation includes creating a user-friendly interface that needs little training. The ultimate goal is to develop a reliable payroll platform that can easily fit into existing admin panel.

1.3 Project Contribution:

The development of this project involved collaboration among four contributors, each responsible for specific components of the system. Their roles and responsibilities are summarized below:

Bidhan Nath (0222220005101027)

- Designed the database structure.
- Developed the employee profile management module.
- Implemented the authentication system.

Aritra Chowdhury (0222220005101024)

- Handled attendance management module
- Developed salary management and payroll functionality.
- Implemented bonus and deduction management features.

Sourav Dev (0222220005101033)

- Designed the admin dashboard interface.
- Developed and managed the admin dashboard functionalities.

Tahsin Ahmed Rafi (0222220005101017)

- Designed the login page.
- Implemented job desk management features.

2. Problem Statement

2.1 Addressing Key System Gaps

Manual payroll processes often lead to mistakes, late payments, and challenges in keeping historical records. Many small and medium-sized businesses also do not have access to affordable payroll tools due to high licensing fees. The Automated Payroll System fills this gap by offering a custom-built, secure, and cost-effective payroll solution that streamlines employee management, attendance tracking, and salary calculations in one integrated platform.

2.2 Challenges in Existing Payroll Systems

Current payroll systems have several shortcomings. Some have complex interfaces that are hard to use. Others do not provide flexibility for creating various salary structures, allowances, or deductions. Additionally, many systems do not integrate attendance, leave management, and payroll. Another major problem is the absence of automated notifications and real-time reports. Employees often need to reach out to HR to get their pay slips or confirm attendance records. The Automated Payroll System addresses these issues with a fully automated workflow. Employees can access their pay slips directly, and administrators can generate detailed reports instantly.

2.3 Efficient Payroll Management with Automation

The proposed system guarantees full automation of payroll processing. The administrator logs into the system, checks attendance and leave data, sets salary configurations, and starts payroll generation. The system then identifies any missing data, calculates gross and net salaries, applies deductions and taxes, and creates digital pay slips which are emailed to employees. This simplified process ensures that salary generation is consistent, clear, and free of errors. It also provides management with a central dashboard to access payroll summaries and financial reports at any time.

3. Objectives

The main goal of this project is to create a fully automated, database-driven payroll management system using Laravel and PHP. The project aims to:

- Automate how employee salaries are calculated based on attendance and leave.
- Implement role-based access for administrators and employees.
- Create accurate pay slips and detailed financial reports.
- Send pay slips automatically.
- Keep a secure database with all payroll transactions for audits and analysis.

In the end, the project seeks to provide an efficient, reliable, and user-friendly payroll system that saves time and reduces errors.

4. Methodology

4.1 Requirement Identification

4.1.1 Literature Review

Several research studies highlight the impact of automation in payroll management. Systems built using frameworks like Laravel provide modularity and data integrity, reducing human dependency and ensuring data accuracy. Existing software like QuickBooks and Zoho Payroll, while powerful, are often expensive and require constant internet connectivity. Therefore, an open-source and customizable solution like the Automated Payroll System offers flexibility, affordability, and efficiency for organizations with varying needs.

4.1.2 Requirement Analysis

To develop this system, specific hardware and software requirements were identified. Software requirements include:

- Operating System: Windows or macOS
- Backend: Laravel 11 using PHP 8.2
- Database: MySQL
- Frontend: PHP with Bootstrap 4
- IDE: Visual Studio Code
- Web Server: Apache (via XAMPP)

4.2 Design Diagrams

The system was designed using several Unified Modeling Language (UML) and structured diagrams to represent its architecture, workflow, and data relationships.

4.2.1 UML Class Diagram

The class diagram illustrates the main entities such as Admin, Employee, Department, Payroll, Pay slip, Tax Rule, Salary Structure, Attendance, and Leave Request. It shows how the administrator manages payroll processes and how the system computes salaries using defined tax rules and salary structures. Each class includes attributes and methods, ensuring object-oriented consistency across the system.

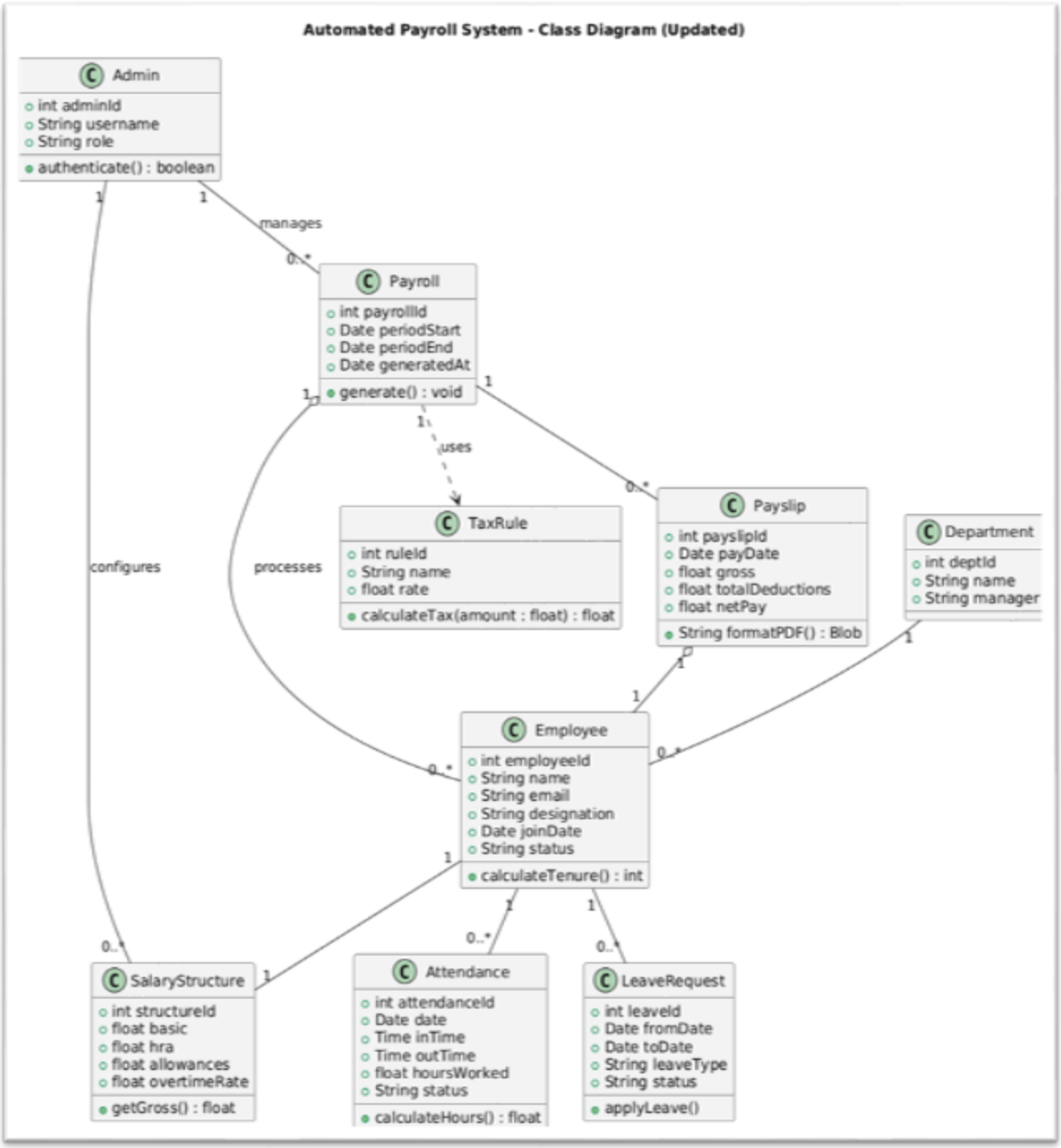


Figure 1: UML CLASS DIAGRAM

4.2.2 Use Case Diagram

The use case diagram explains the interactions between the system and its users. Administrators can perform operations such as managing employees, generating payroll, configuring salary structures, and sending pay slip emails. Employees can log in, view their

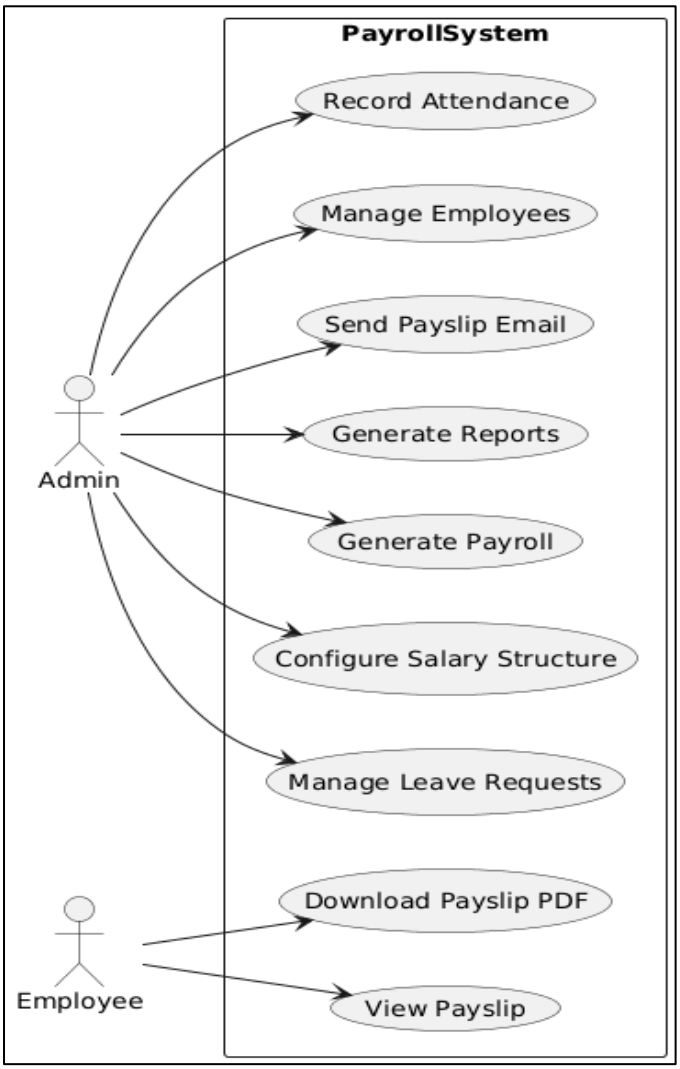


Figure 2: USE CASE DIAGRAM

4.2.3 Entity Relationship Diagram (ERD)

The ERD shows the relational structure of the database. Entities like Employee, Payroll, Pay slip, and Attendance are connected through primary and foreign keys. Each employee record is linked to multiple attendance and leave entries, and each payroll period can generate multiple pay slips. This structure guarantees efficient normalization and consistency within the system's data management.

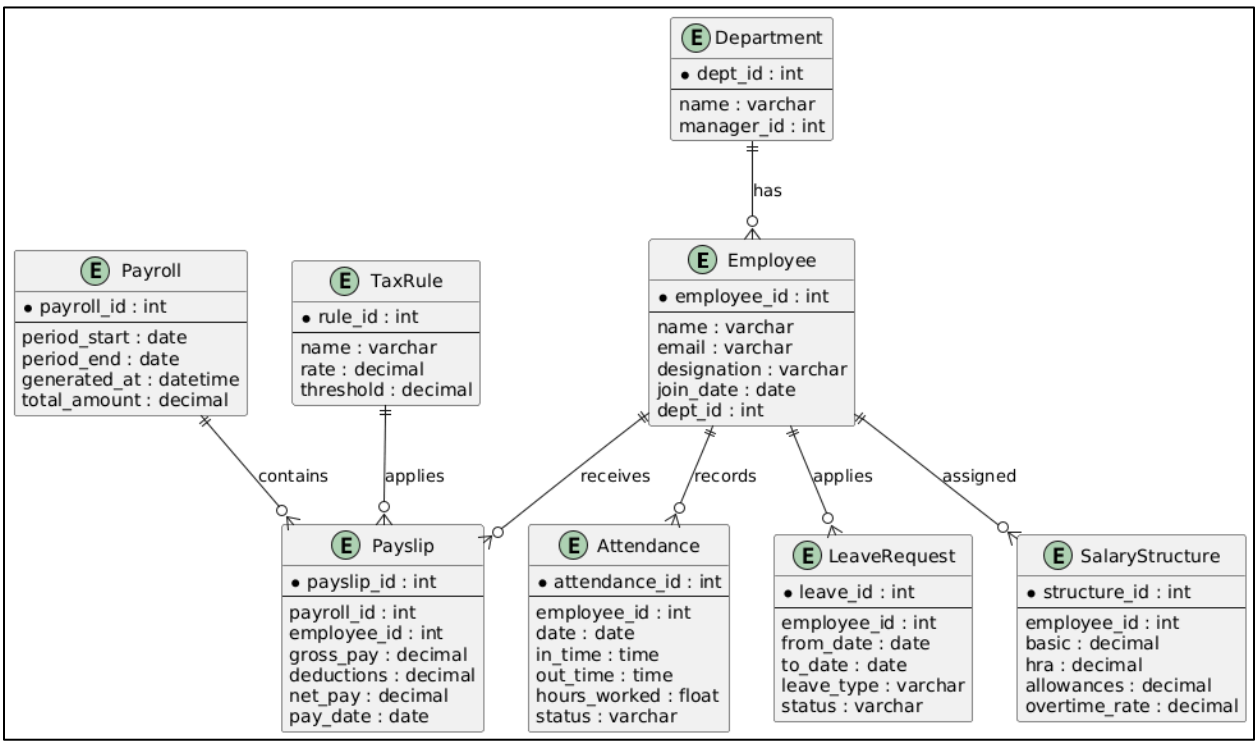


Figure 3: Entity Relationship Diagram (ERD)

4.2.4 Activity Diagram

The activity diagram depicts the flow of operations during payroll generation. The administrator logs into the system, verifies attendance, configures salary and tax rules, and initiates payroll. The system checks for any missing data; if found, the admin is notified. Otherwise, the process continues with gross pay calculation, deduction of taxes, and pay slip generation. Finally, the system stores payroll data in the database and emails the pay slips to employees.

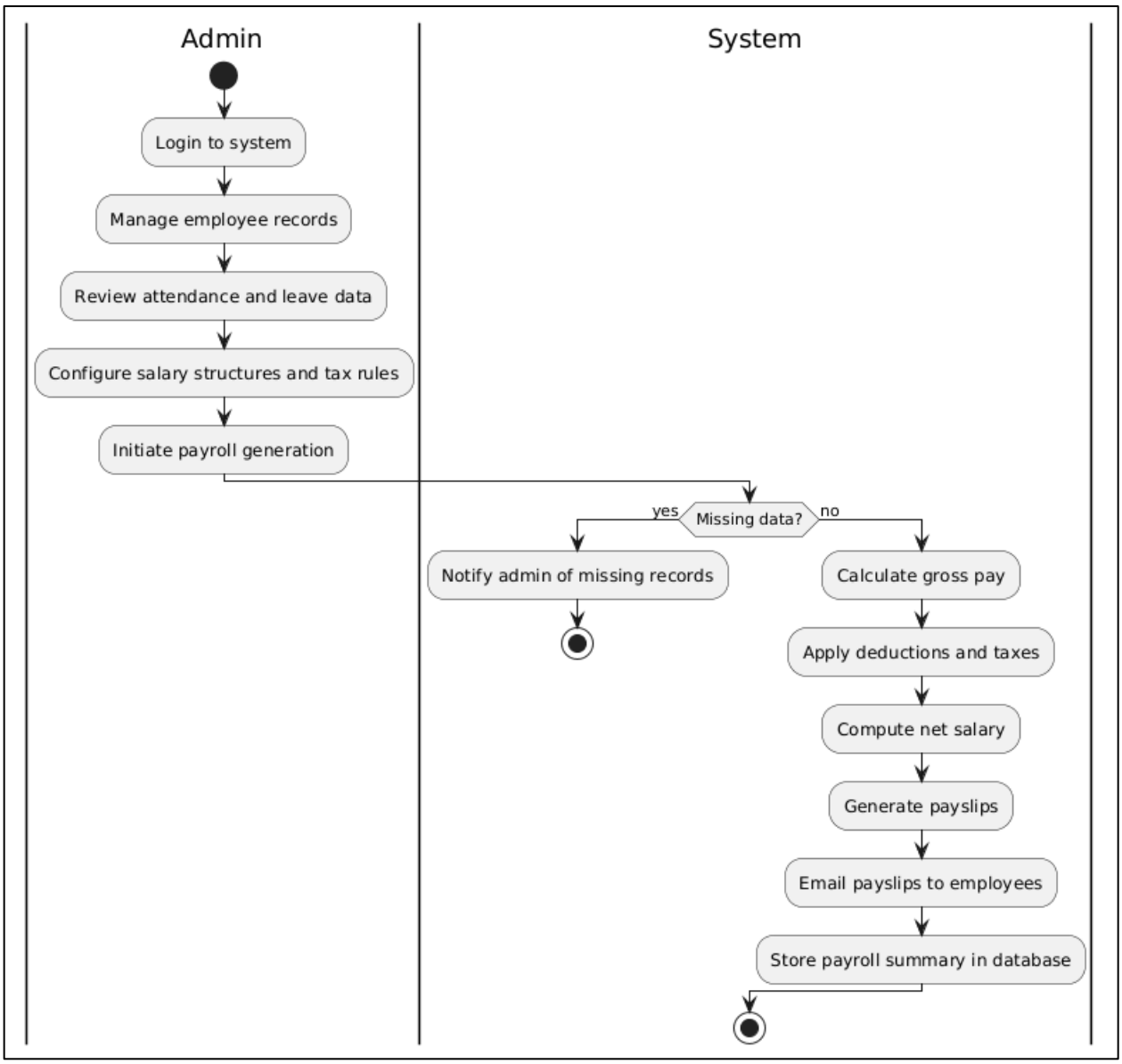


Figure 4: ACTIVITY DIAGRAM

4.2.5 Sequence Diagram

The sequence diagram details the dynamic interaction between system components. When the admin initiates payroll generation, the Payroll System communicates with the Payroll Database to fetch employee and salary data. It then processes computations for each employee, saves the pay slip record, and triggers the Email Service to deliver pay slips. The system then confirms completion and displays a payroll summary to the admin.

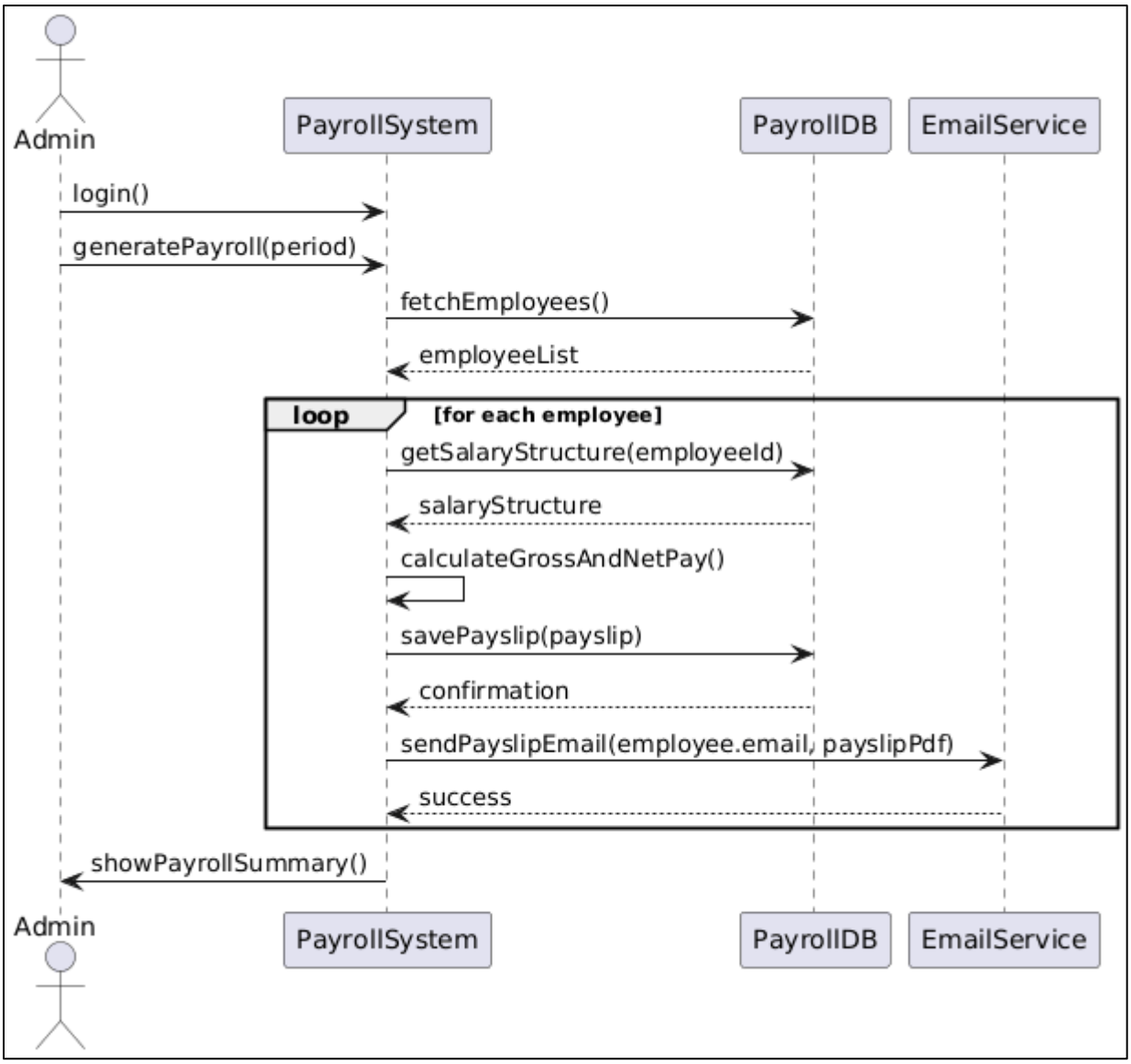


Figure 5: Sequence Diagram for Payroll Generation

4.2.6 Data Flow Diagram

The DFD provides an overview of data movement within the system. It illustrates how the admin manages employee information, attendance, and leave data, which is then stored in the Payroll Database. The system reads and processes this information during payroll generation and uses an integrated Email Service to notify employees with pay slips. This modular flow ensures smooth communication between components and eliminates data redundancy.

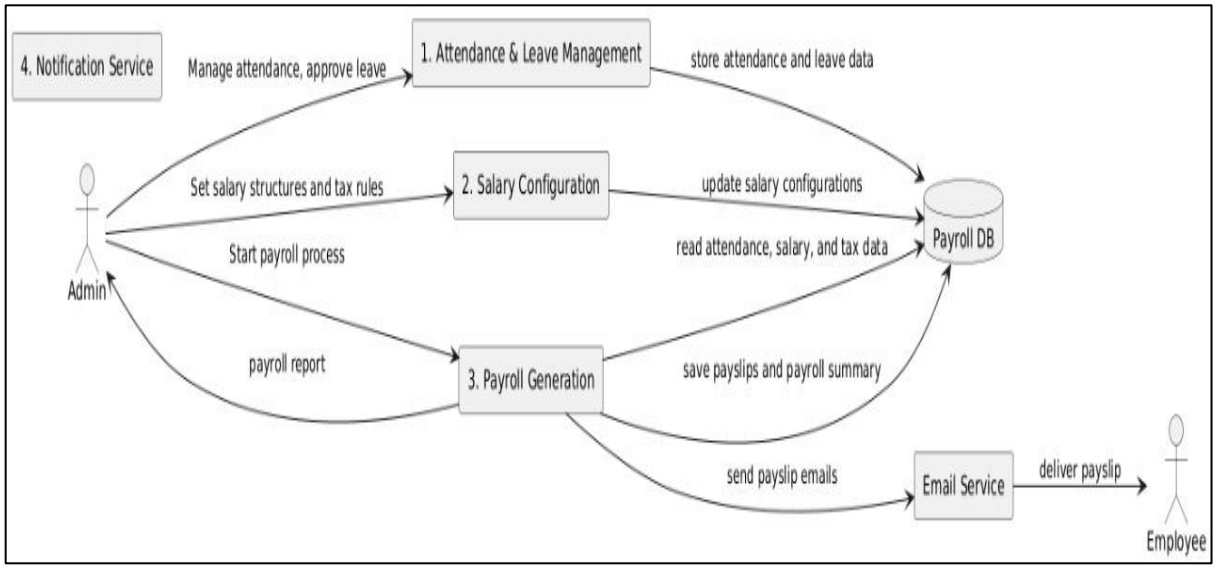


Figure 6: DATA FLOW DIAGRAM

4.2.4 Activity Diagram

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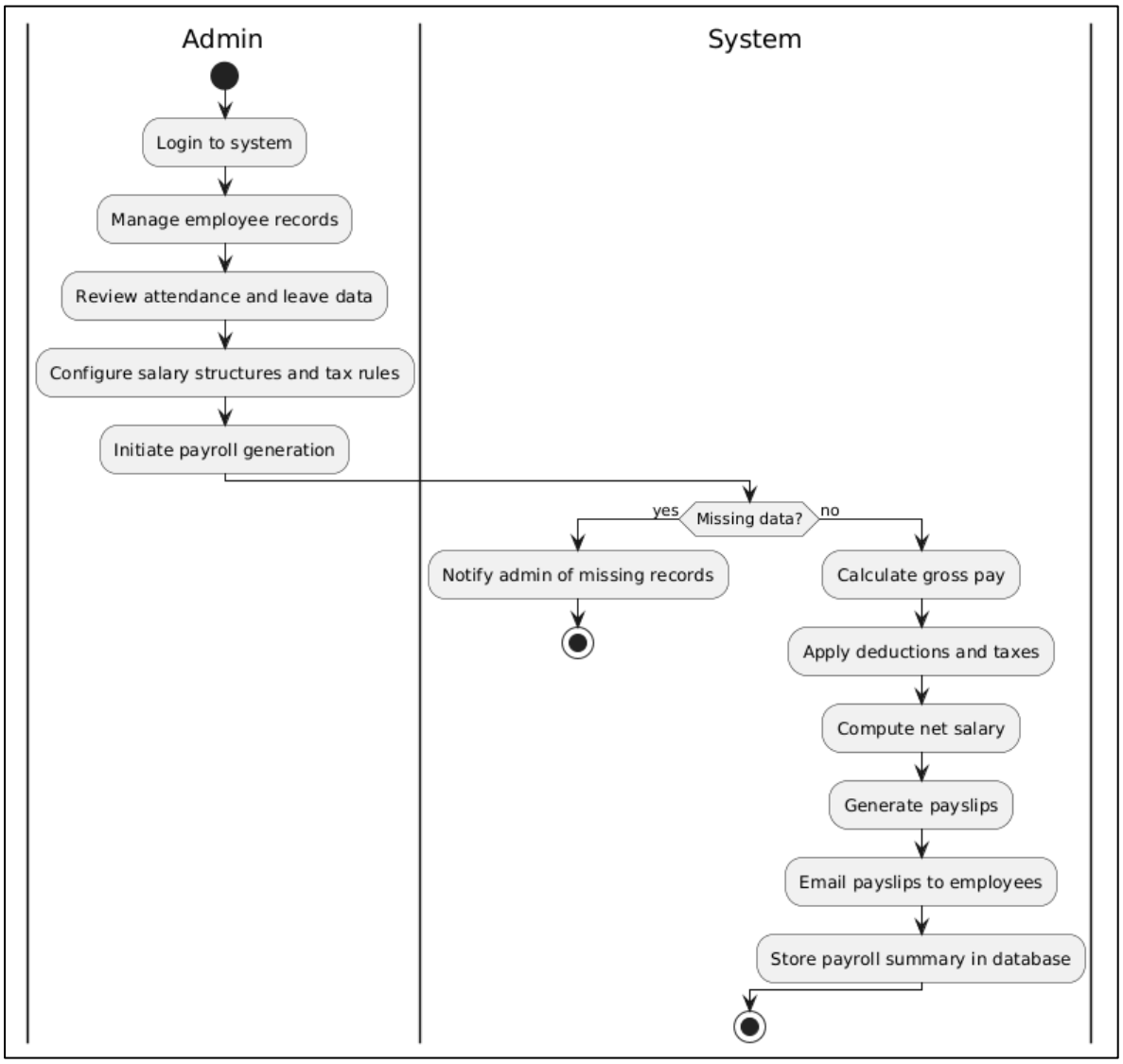
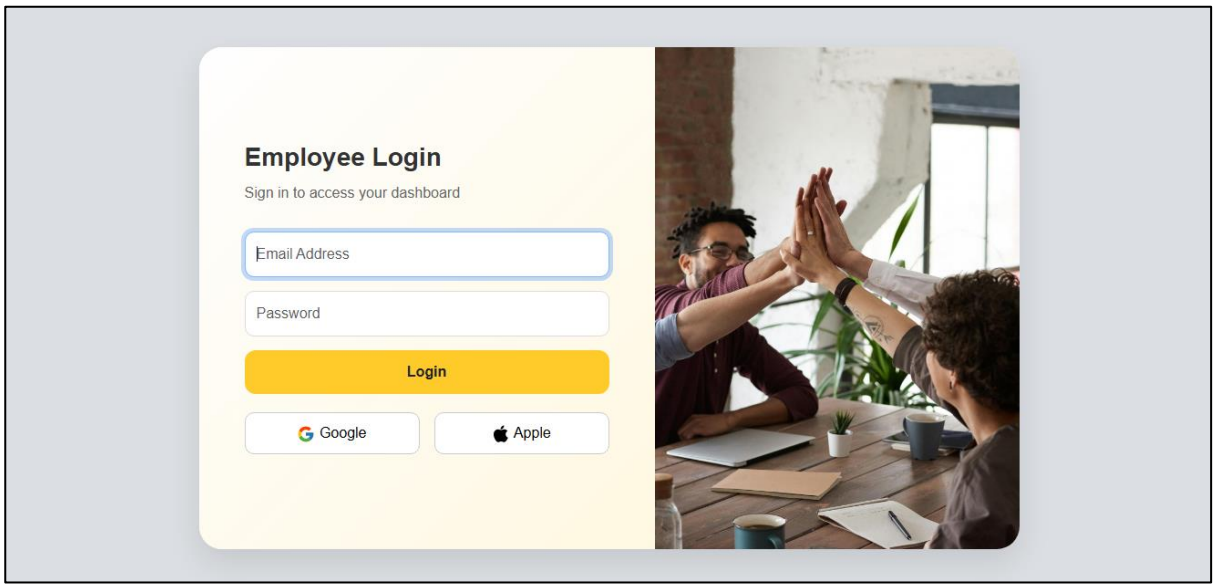


Figure 7: ACTIVITY DIAGRAM

5. Implementation

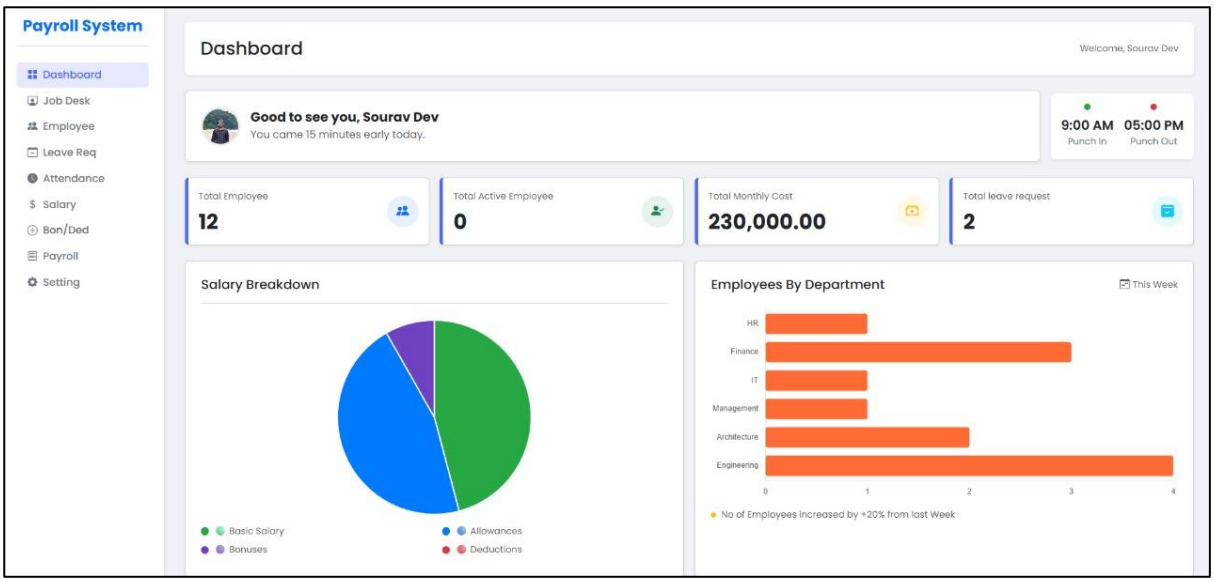
5.1 Employee Login Portal:

This screen serves as the secure login portal for employees to access the payroll system. It features a clean and modern design with a standard email and password login form. For added convenience and security, the page also provides social login options, allowing users to authenticate quickly using their Google or Apple accounts.



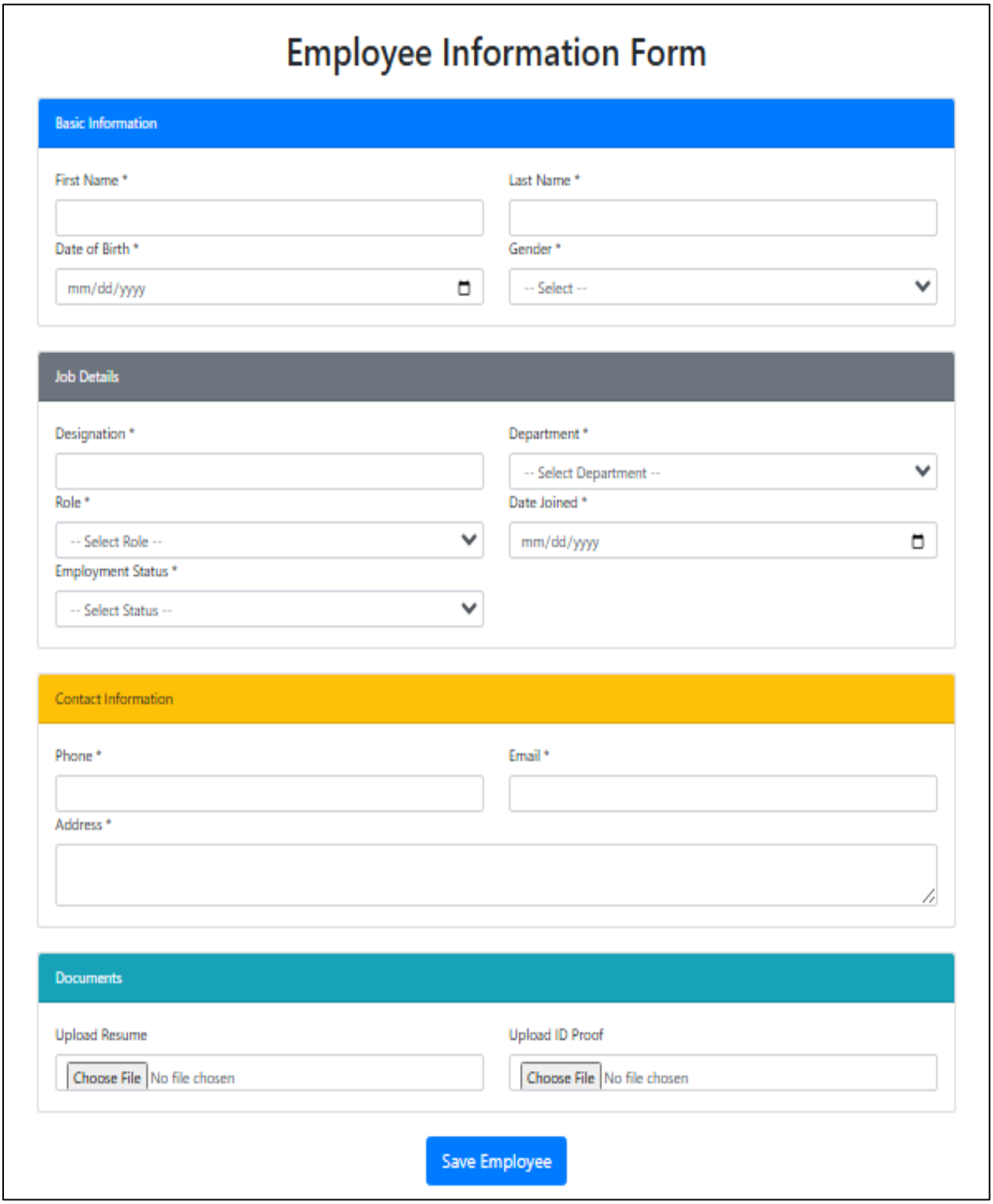
5.2 Admin Dashboard & Analytics:

This is the main administrator dashboard, providing a high-level, command-center view of the entire organization. It features key performance indicators (KPIs) in summary cards, including the total number of employees, total monthly payroll cost, and pending leave requests. The dashboard also includes data visualizations, such as a pie chart for the company-wide salary breakdown and a bar chart showing the number of employees by department, giving administrators immediate insight into workforce distribution and financials.



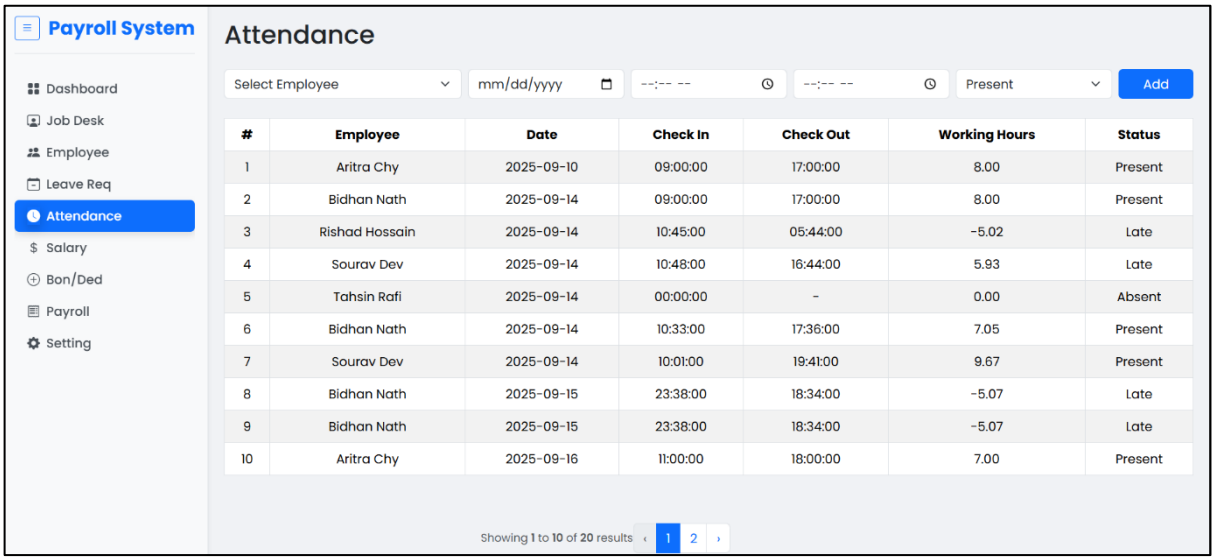
5.3 New Employee Onboarding Form:

This is the new employee onboarding form, used by administrators to add new staff members to the system. The form is logically organized into distinct sections—Basic Information, Job Details, Contact Information, and Documents—to streamline the data entry process. It captures all essential details, from personal and professional information to necessary document uploads like resumes and ID proofs, before saving the complete record to the employee database.



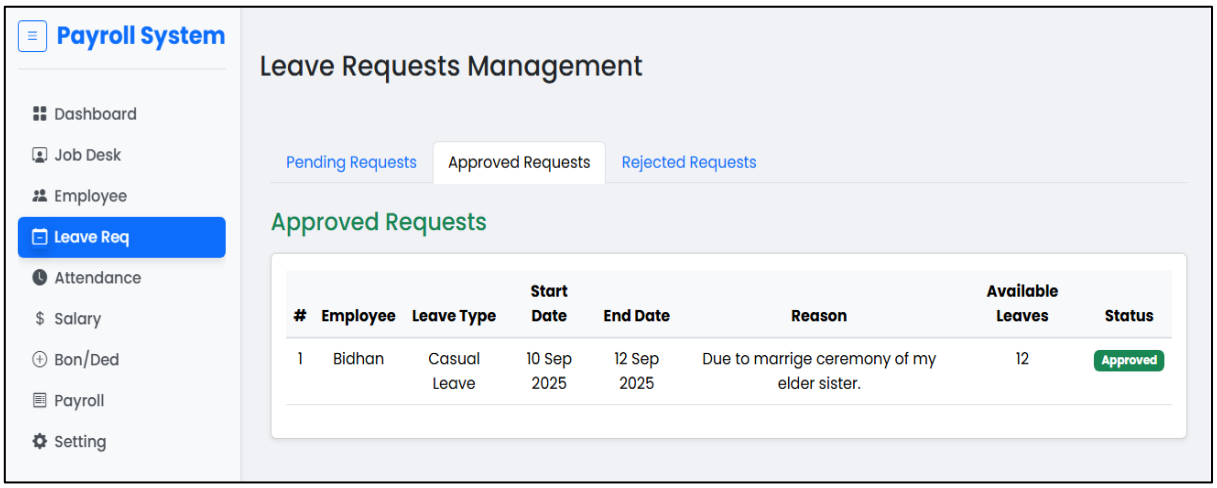
5.4 Employee Attendance Log & Management:

The attendance management page provides a detailed log of employee check-in and check-out times. Administrators can filter records by employee and date or manually add new entries as needed. The system automatically calculates the total working hours for each day and displays a clear status, such as 'Present' or 'Absent', making it easy to monitor daily workforce presence and punctuality.



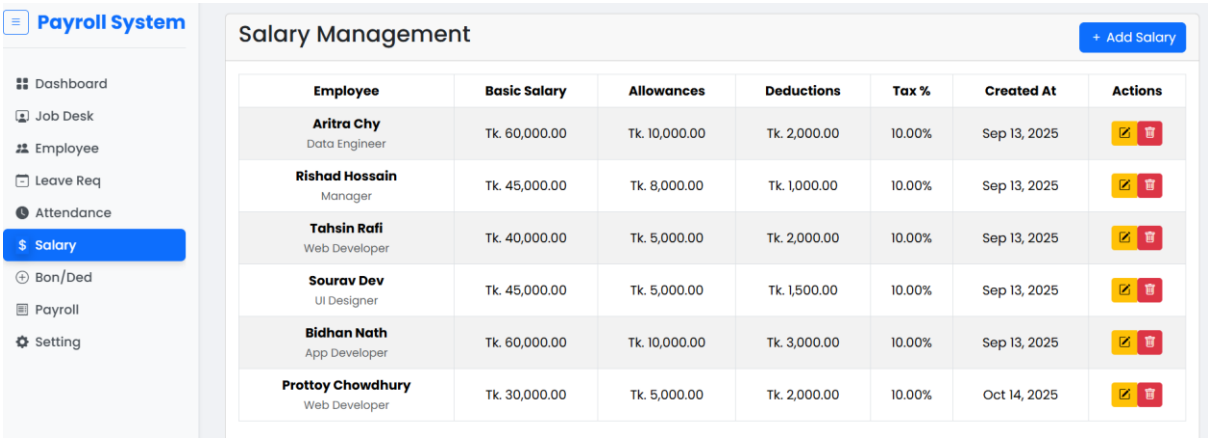
5.5 Administrator Leave Request Dashboard

This is the administrator's dashboard for overseeing all employee leave applications. The interface uses a tabbed navigation system to efficiently organize requests into 'Pending', 'Approved', and 'Rejected' categories, allowing for streamlined workflow management. Each entry in the list provides the administrator with all necessary details, including the employee's name, leave type, duration, and reason, to make an informed decision.



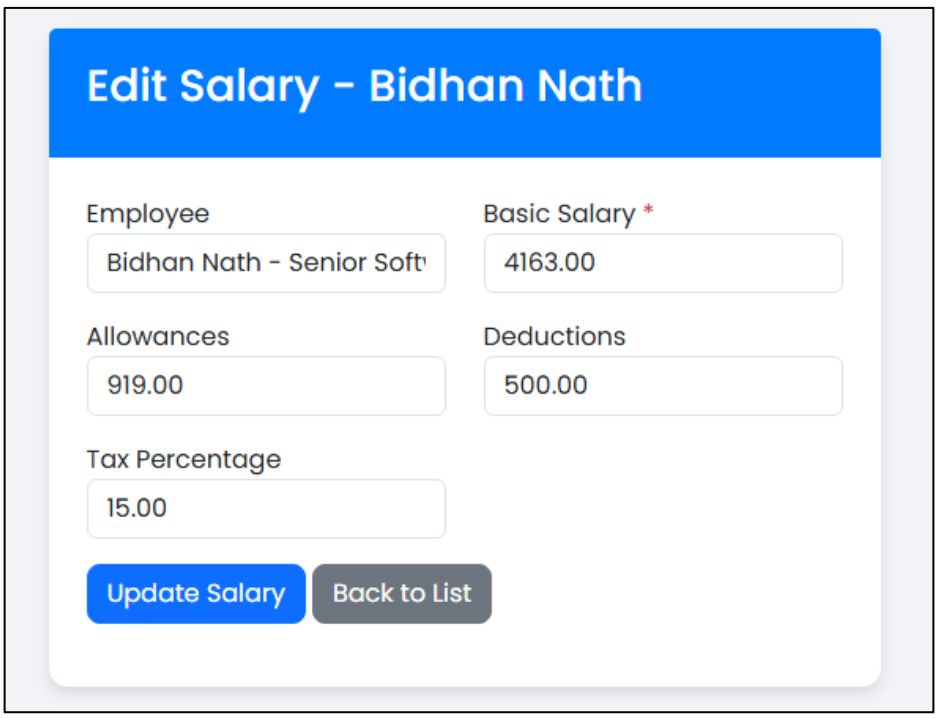
5.6 Salary Structure Management:

The Salary Management screen is where administrators configure and oversee the compensation packages for each employee. The system provides a detailed breakdown of each employee's salary, including basic pay, allowances, deductions, and tax percentages. Administrators have full control to add new salary structures or modify and delete existing records using the action buttons.



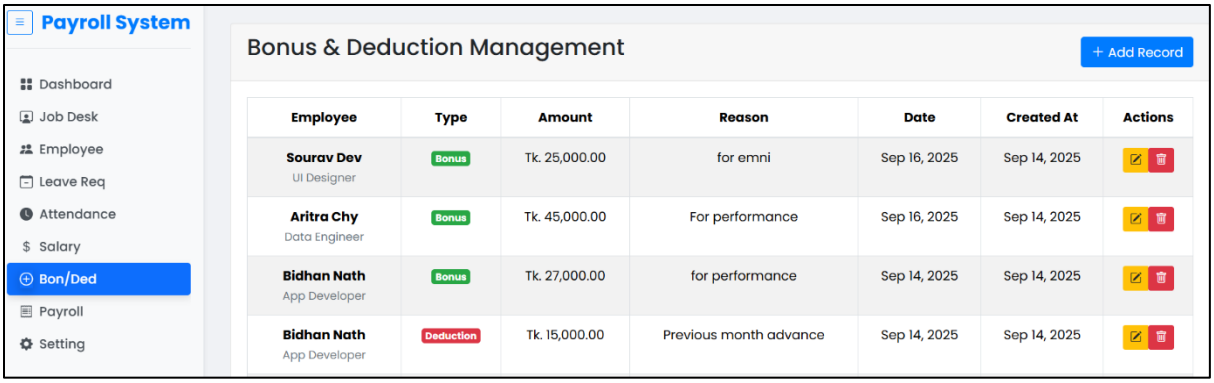
5.7 Edit Employee Salary Form:

This form allows an administrator to edit the salary structure for a specific employee. The fields are pre-populated with the current financial data, including the employee's basic salary, allowances, deductions, and tax percentage. The administrator can make necessary adjustments to any of these components and then update the employee's compensation record in the system.



5.8 Bonus & Deduction Management:

This page is used to manage ad-hoc financial adjustments for employees, such as performance bonuses or penalties. Each record clearly specifies the employee, transaction type, amount, reason, and effective date. The system uses color-coded badges to visually distinguish between a 'Bonus' and a 'Deduction', and administrators can add, edit, or remove records as needed.



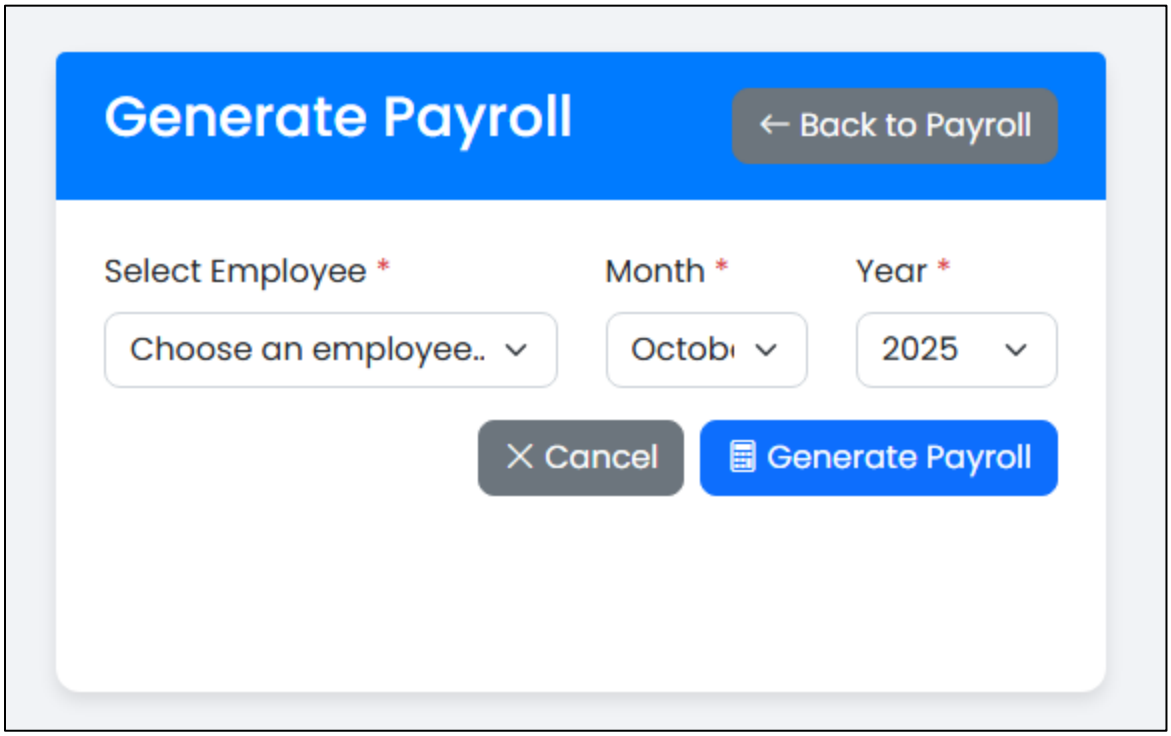
5.9 Payroll Management & History

The Payroll Management screen serves as the primary hub for processing and reviewing employee pay slips. Administrators can initiate new payroll runs using the 'Generate Payroll' button. The main table provides a historical log of all processed payrolls, detailing the gross salary, total deductions, bonuses, and the final net salary for each pay period. Action buttons for each record allow for viewing the detailed pay slip or downloading it directly.



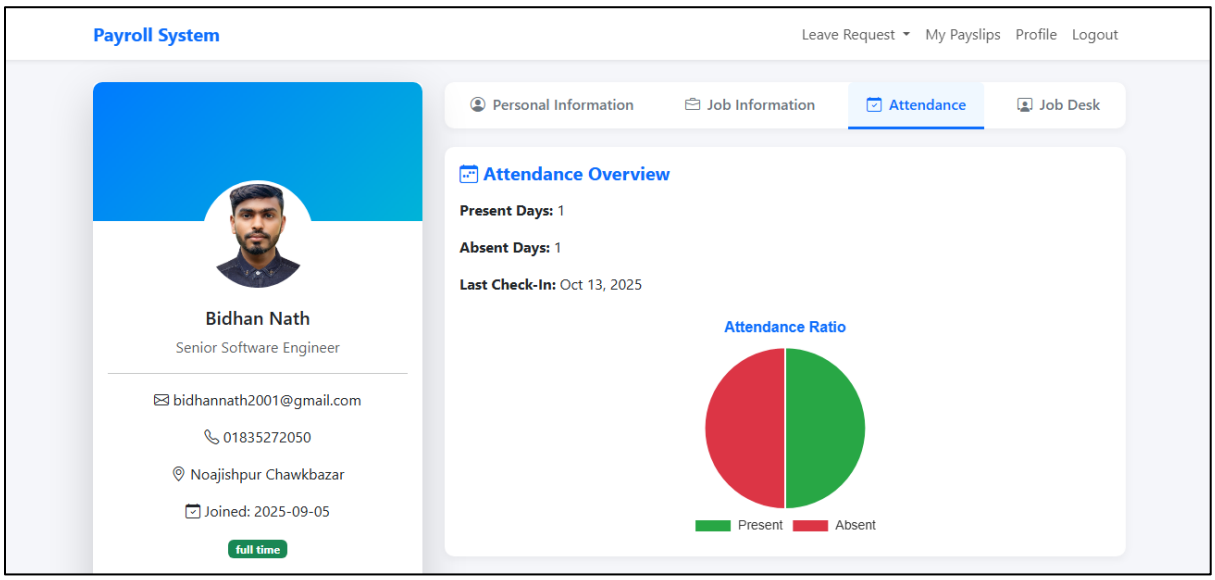
5.10 Payroll Generation Form:

This form is the starting point for the payroll generation process. The administrator selects a specific employee and the relevant month and year for the pay period. Upon clicking 'Generate Payroll', the system processes all associated data—including salary, attendance, bonuses, and deductions—to calculate and create a new pays lip record for the selected employee.



5.11 Employee Profile & Attendance Overview:

This screen displays the detailed profile for an individual employee. The left panel shows static personal and contact information, while the main content area is organized into tabs. The active 'Attendance' tab provides the employee with a clear overview of their attendance record, including a summary of present and absent days and a pie chart that visually represents their attendance ratio.



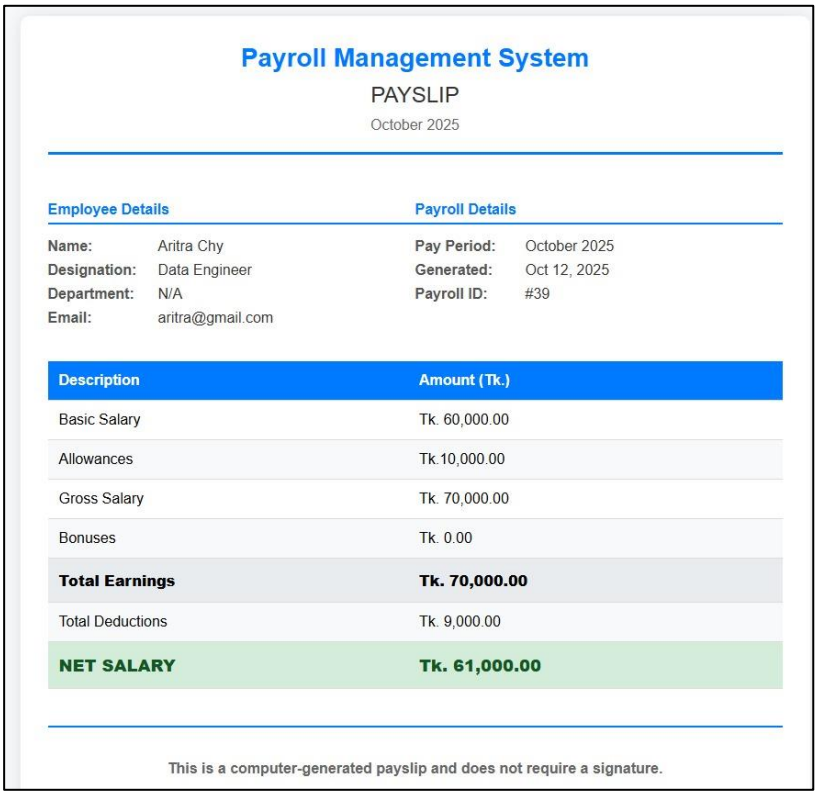
5.12 Leave Request Management & Status Tracking:

This screen provides a centralized view for managing and reviewing all employee leave requests. The list details essential information for each application, including the leave type, duration, and reason. A clear, color-coded status badge for each entry—such as 'Approved' or 'Rejected'—allows for quick and easy tracking of the outcome of each request.



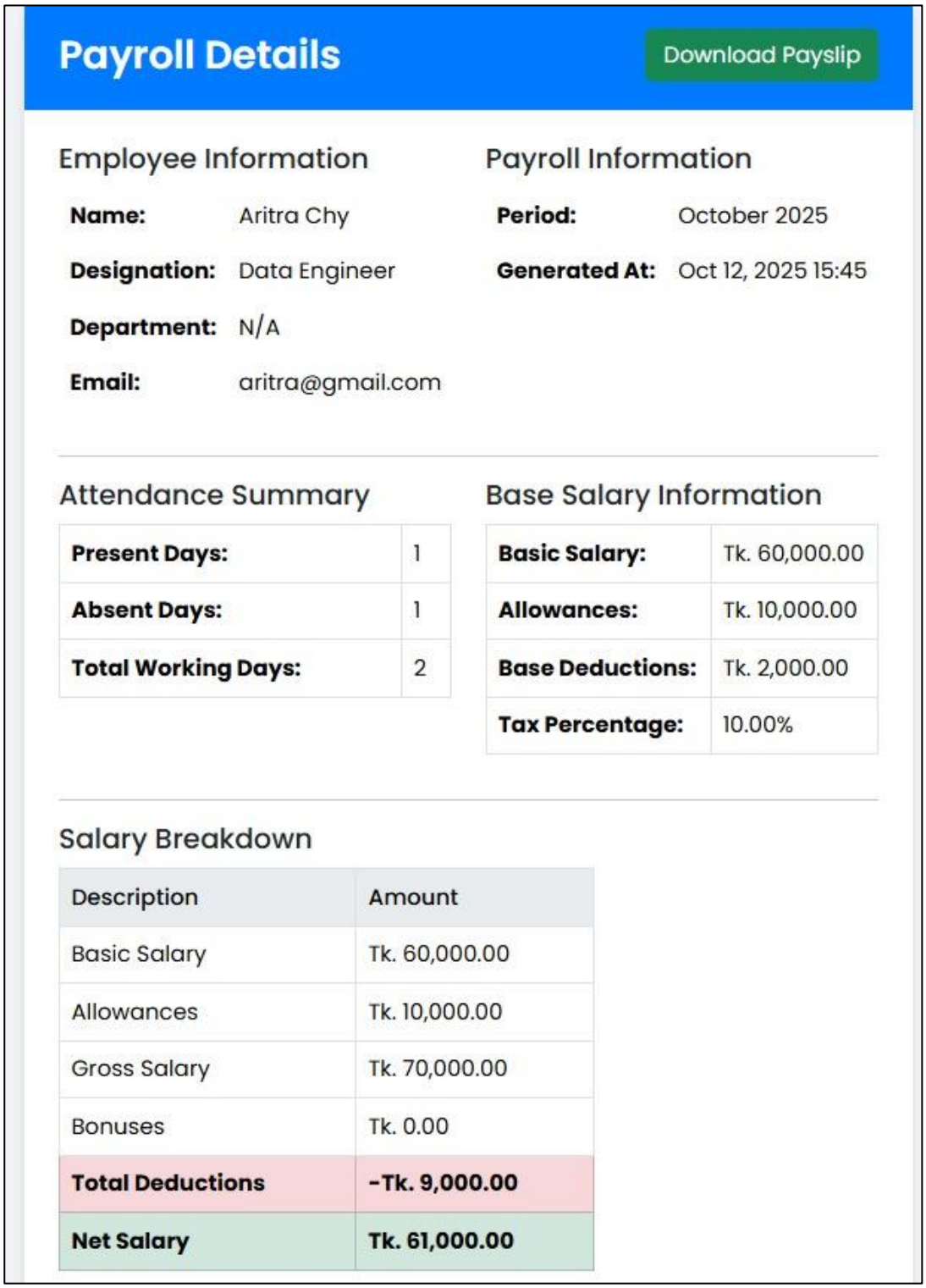
5.13 Downloadable Employee Pay slip:

This image shows the final, downloadable pay slip document generated by the system. It's formatted as an official record, presenting employee and payroll details at the top, followed by a clear, itemized financial breakdown. The main table transparently calculates the employee's pay, summarizing the path from basic salary to total earnings and the final net salary, providing a formal statement for the employee's records.



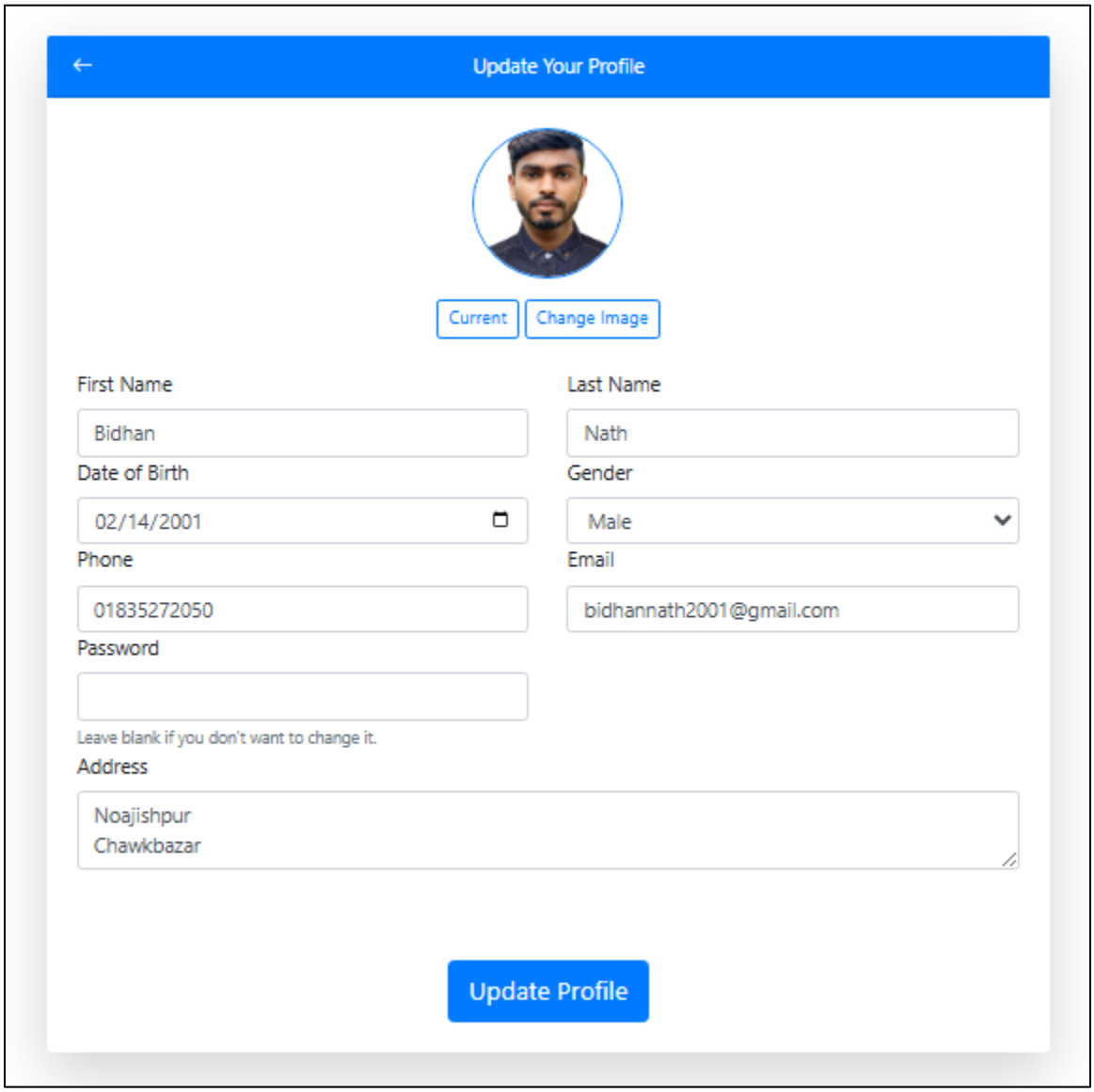
5.14 Detailed Employee Pay slip:

This screen presents the detailed pays lip for an individual employee for a specific pay period. The information is organized into clear sections, including employee details, an attendance summary, and base salary components. The final 'Salary Breakdown' table provides a transparent calculation, starting from the basic salary and culminating in the final net salary, with deductions clearly highlighted. Users have the option to download a copy of the pay slip for their records.



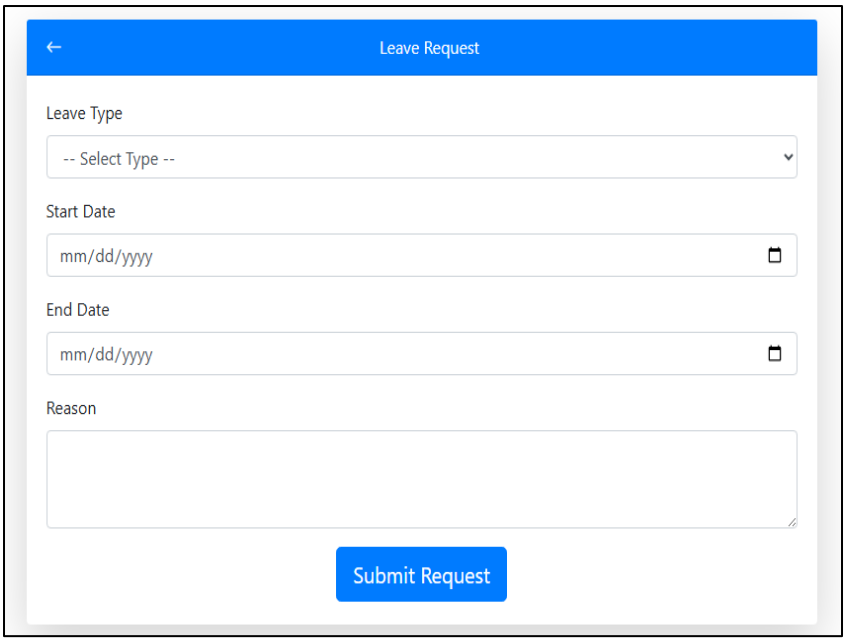
5.15 Employee Profile Update Form:

This form provides a self-service interface for employees to update their personal information. Users can modify their name, contact details, and address, as well as change their profile picture. The form is pre-populated with the user's current data for easy editing and includes an option to update their password securely.



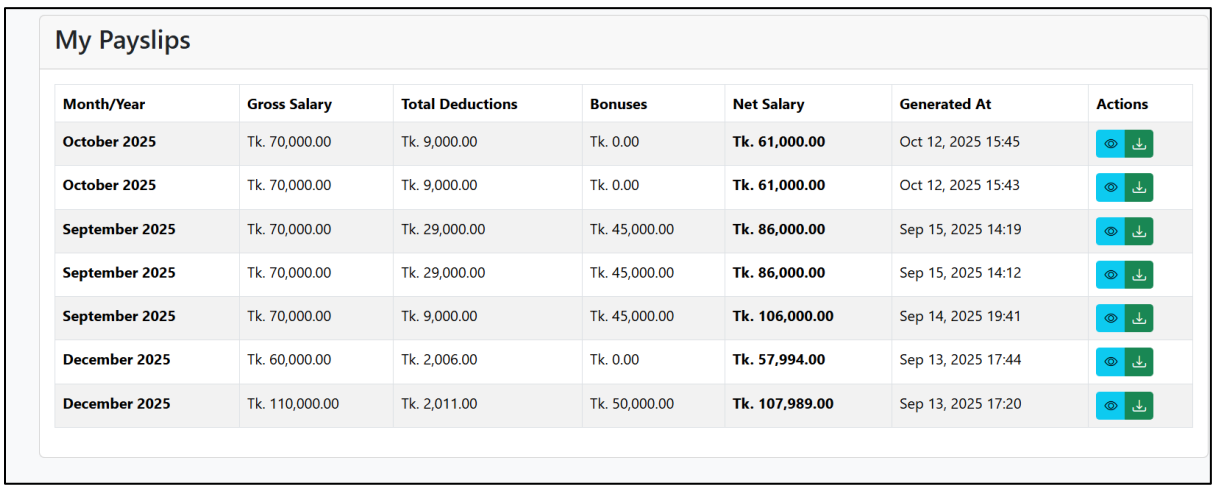
5.16 Employee Leave Request Form:

This page provides a simple and intuitive form for employees to submit leave requests. The user must select the type of leave, specify the start and end dates of their absence, and provide a brief reason for the request. Submitting the form sends the application to the relevant manager or administrator for review and approval.



5.17 Employee Pay slip History:

The 'My Pay slips' page provides employees with a personal, historical archive of their salary statements. For each pay period, the system displays a clear summary including gross salary, total deductions, bonuses, and the final net salary. Employees can use the action buttons to either view a detailed breakdown of the pay slip online or download a copy for their personal records.



6. Conclusion

The Automated Payroll System effectively meets its main goal of making payroll management digital and automatic. This improves accuracy, transparency, and efficiency within an organization. By combining modules for employee management, attendance tracking, leave handling, and payroll generation, the system offers a complete solution for both administrators and employees. Using Laravel for the backend and PHP for the frontend creates a secure, scalable, and user-friendly environment that reduces human errors and administrative burdens.

This system overcomes the shortcomings of traditional payroll processes by removing manual calculations, cutting delays, and keeping reliable financial records. Features like automated pay slip generation, real-time reporting, and database-driven auditing ensure that payroll operations are accurate and meet regulations. The organized structure, shown through UML diagrams, ERD, and DFD models, highlights a solid grasp of software design and data flow within the system.

In summary, the Automated Payroll System is an efficient, sturdy, and flexible platform suitable for organizations of various sizes. It simplifies HR and finance tasks while fostering a paperless and transparent workplace culture. Future improvements might include mobile app integration, enhanced analytics dashboards, and AI-driven insights to further boost payroll forecasting and workforce management.